



(An Autonomous Institution affiliated to VTU, Belagavi)

Supplementary Examinations - December 2025

20CS406 / 19CS404 – PRINCIPLES & PRACTICES OF SOFTWARE ENGINEERING

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Marks	BT*	CO*	PO*
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|----|----|--|----|-----|---|---|
| 1. | a) | Explain the main activities in the software design process and the outputs of these activities. | 10 | L*2 | 1 | 2 |
| | b) | Explain the goals of software engineering. | 10 | L2 | 1 | 1 |
| 2. | a) | Describe the good properties of SRS documents. | 10 | L2 | 1 | 1 |
| | b) | With a neat diagram explain requirement engineering process model. | 10 | L2 | 1 | 1 |
| 3. | a) | Explain the fundamental development activities involved in waterfall model. | 10 | L2 | 2 | 1 |
| | b) | Explain Incremental development process model with a neat block diagram. List its benefits and problems. | 10 | L2 | 2 | 1 |

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|----|----|---|----|----|---|---|
| 4. | a) | Define System modeling? Explain the different perspective that the system model developed. | 10 | L2 | 2 | 1 |
| | b) | Illustrate the sequence diagram with an example to view patient information. | 10 | L2 | 2 | 1 |
| 5. | a) | With the help of neat state diagram, illustrate the working of microwave oven. | 10 | L2 | 3 | 1 |
| | b) | Draw and explain use case modeling and sequence diagram for patient information system. | 10 | L2 | 3 | 1 |
| 6. | a) | Define architectural design? List out architectural design decisions. Explain architecture and system characteristics. | 10 | L2 | 3 | 1 |
| | b) | Define architectural Patterns? Explain the organization of MVC architecture with an example of web application architecture mention its advantages and disadvantages. | 10 | L2 | 4 | 1 |

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|----|----|---|----|----|---|---|
| 7. | a) | Define Risk management? Explain in detail risk management process with block diagram. | 10 | L2 | 5 | 1 |
| | b) | Explain with the block diagram project planning process in plan driven development. | 10 | L2 | 5 | 1 |
| 8. | a) | Give an example of common project, product and business risks. | 10 | L2 | 5 | 1 |
| | b) | Briefly explain the types of estimation techniques. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
